

JOB SAFETY ANALYSIS – MARINE (FLOATING BARGE) DRILLING

SCOPE	This Job Safety Analysis (JSA) addresses the risks and hazards involved with drilling from a floating barge in a marine environment. The purpose is to outline specific hazards associated with the safe operation of equipment from a floating barge platform and the control measures used to eliminate or minimise risk to operators, assistants and others. Specific task analysis is identified in this document where hazard, risk and controls are specified.
LIMITATIONS	This document is to be read in conjunction with McMillan's Health & Safety Manual, Health and Safety at Work Act 2015 (HSWA), NZS 4411:2001 and relevant Codes of Practice.
EQUIPMENT DETAILS	Barge, marine support vessel (if required), drill rig, tooling, ancillary equipment
PERSONNEL	All personnel operating rigs are fully trained and competent in the use of the equipment listed above. Drivers Licences and Site Safe Passports (where appropriate) are carried at all times by staff.
STANDARD PPE	McMillan staff are issued with standard personal protective equipment and trained in its use. This includes, however is not limited to the following; • SAFETY HELMETS – to be worn at all times. • PERSONAL FLOTATION DEVICES (PFDs) – to be worn at all times when over water. • HIGH VISIBILITY GARMENTS – to be worn at all times. • SAFETY FOOTWEAR – to be worn at all times. • SAFETY GLASSES – to be worn as necessary. • EAR PROTECTION – is to be worn during operation of the rig. • GLOVES – are provided and are to be worn as necessary.
PREPARATORY WORK	All sites must be checked for hazards prior to any work commencing. This includes, however is not limited to the following; • General Housekeeping • Sub Surface Services (buried) • Identifying services by hand clearing where appropriate. • Overhead Services (i.e. Power Lines). No rig will be operated within 4m of overhead power cables. All hazards caused by other contractors who may be working on site must be assessed and notified prior to McMillan's undertaking any work.
GENERAL SAFETY NOTES	All personnel, including geotechnical consultants shall comply with minimum PPE requirements of Safety Helmets and High Visibility outer garments are worn (where required) when within 10m of a drilling rig, operational or not. The operator of the rig is the person responsible for the immediate drilling site. Standard PPE comprises of Safety Boots, Glasses, Overalls, Hard Hat and Grade 5 Ear Muffs.
RESIDUAL RISKS	Staff who are training will be closely supervised at all times to minimise any risk to themselves and others. All personnel directly involved must pay full attention to the task at hand. Weather conditions can have a significant effect on drilling. They can make surfaces slippery and reduce visibility. If drilling cannot be postponed all care must be taken.

TASK SPECIFIC JSA – MARINE (FLOATING BARGE) DRILLING

HAZARD & RISK SCORE	HARM	INITIAL RISK	HAZARD CONTROLS	ELIMINATE	MINIMISE	RESIDUAL RISK
Potentially Hazardous Task / Action / Situation	Potential Harm	LOW MED HIGH	Control Steps			LOW MED HIGH
SITE SPECIFIC HAZARDS – when there is a change in the work process, when there is a change in personnel or when a new hazard is introduced e.g. property, plant, or environmental hazards these need to be added to the SSSP / Job Card / Daily Pre-Shift Form						
				☐	☐	
				☐	☐	
				☐	☐	
SITE CHECK / ESTABLISHMENT						
Pre-site check / barge inspection	General hazards & safety	●	Complete the SSSP and Daily Pre-Shift (PSI). Inspect barge deck for housekeeping, slip/trip hazards, handrail integrity, emergency equipment, and fuel/hydraulic systems. Establish exclusion zones. Barge to be secured prior to operations commencing.	☐	✓	●
Barge deck housekeeping	Trip hazards, fracture, strain, sprain, abrasion	●	Keep barge deck clean, clear and tidy at all times. All equipment to be stowed or secured when not in use. Offsider to maintain clear deck throughout operations.	☐	✓	●
Unauthorised access / public interaction	Injury to persons from drilling or vessel operations	●	Establish and maintain exclusion zones around drilling and lifting operations. All visitors to be inducted prior to accessing the barge. Signage posted at access points.	✓	☐	●
Marine vessel traffic / exclusion zones	Collision; injury from vessel movement; distraction to crew	●	Maintain exclusion zones around barge. Communicate with other marine users as required. Barge navigation lighting operational when at anchor. Harbour Master to be notified as required by project SSSP.	✓	☐	●
MARINE TRANSFERS & ACCESS / EGRESS						
Transfer between support vessel and barge	Falling overboard; crush injuries; drowning	●	Transfers are only to be undertaken using stable, agreed arrangements under direction of the barge master or vessel operator. All personnel to wear PFDs at all times. Three points of contact maintained during all transfers — hands must be free when stepping between vessels. Personnel not directly involved are to stand clear. Transfers suspended during adverse weather, swell, wake, or poor visibility.	☐	✓	●
Access / egress on barge deck (wet surfaces)	Slips, trips, falls overboard	●	Non-slip surfaces maintained on barge deck. Handrails used at all times adjacent to open water. PFDs worn. Deck kept clear of trip hazards at all times. Personnel to remain within designated work areas.	☐	✓	●
Working adjacent to open water	Drowning; hypothermia	●	PFDs worn at all times by all personnel on the barge or support vessel. Guardrails and handrails maintained where provided. Man-overboard rescue equipment (life rings, throw lines) available on barge at all times. Man-overboard procedures communicated to all crew and tested at commencement of marine operations. Non-essential personnel excluded from deck areas during high-risk activities.	☐	✓	●

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BARGE STABILITY & RIG SETUP						
Barge movement during drilling	Rig instability; personal injury; equipment damage		Barge positioned and secured prior to drilling using spuds or anchoring arrangements as determined by the barge operator. Spuds fully deployed and holding capacity confirmed before drilling commences. Barge movement monitored throughout operations — drilling suspended if excessive movement occurs due to weather, tide, or wake.	☐	✓	
Rig start-up and pre-start checks	Serious injuries from uncontrolled plant		Pre-start checklist completed and kept with the job card. All controls and levers in neutral. Hydraulic hoses and cables checked for condition. Emergency shutdown tested and confirmed operable prior to start.	☐	✓	
Loading / securing rig and equipment onto barge	Dropped loads; crush injuries; equipment damage; barge instability		Lifts performed by competent operators and trained dogmen using certified gear. Lift plans in place for all crane lifts. No work under suspended loads. Tag lines not to be used in a manner that requires personnel to position themselves under loads. Equipment secured and load distribution checked prior to departure from wharf.	☐	✓	
Spud deployment and retrieval	Hydraulic hose failure; environmental spill; personal injury		Hydraulic hoses checked as part of pre-start inspection. Spill kit available at all times. Personnel not involved in spud operations to move to a safe location as directed by the barge master. If spuds fail to hold, barge to be repositioned and spuds redeployed — Drilling Supervisor to be contacted if holding cannot be confirmed.	☐	✓	
Rig alignment and stabilisation on barge	Rig instability; falling; equipment damage		Rig to be stable and secured to barge deck before drilling commences. Rig chained or positively restrained using rated and compliant load restraints. Tool string alignment confirmed. All loose equipment secured before any barge relocation or drilling operations.	☐	✓	
RIG OPERATION						
Proximity to excessive noise	Impaired hearing – short term or permanent		PPE must be utilised – earplugs or grade 5 earmuffs. Monitoring of condition and utilisation – ongoing. If PPE is compromised or damaged it must be replaced immediately.	☐	✓	
Loading drill string	Strains, sprains, crushing		Load drill string using winch line with hook where practicable. Elevate top joint and rotate head to connect. No manual handling of drill string whilst rotation is applied. Minimum 60 cm exclusion from rotating string enforced. 1.5 m rods used where applicable. Correct manual handling techniques applied — lift using knees, do not twist and lift simultaneously, keep loads close to the body.	☐	✓	
Pulling rods	Strains/sprains, cuts, abrasion		Correct lifting and pulling techniques implemented. Use lifting aids and hydraulic cables wherever possible. Two-person lift where mechanical aids are not practicable.	☐	✓	
Making up joints	Strains/sprains, cuts, abrasion, puncture		Top joint clamped with lower hydraulic jaws. Rotational force applied by top drive or hydraulic jaws to make up joints. Interlocked spin guard to be open when making up joints manually — this prevents	☐	✓	

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			powered rotation. Rods nipped up by hand then tightened with wrench. No combination of manual and mechanical methods. Driller to show hands before any manual handling. Class 2 cut-resistant gloves to be worn when handling rods.			
Breaking out joints	Strains/sprains, cuts, abrasion, puncture		Top joint of bottom rod clamped with lower hydraulic jaws. Rotational force applied by upper jaws. Interlocked spin guard to be open when breaking out joints manually — this prevents powered rotation. Even pressure applied — no jarring motion. Rig to be shut down if manual breaking out is required. Driller to show hands before manual handling. Class 2 cut-resistant gloves to be worn.	☐	✓	
Drilling	Entanglement; pinch points; crush injuries		Assistant to maintain minimum 60 cm distance from drill string during rotation (delineated by foot clamps). Assistant may only enter zone when string is stationary and operator takes hands off all controls. Interlocked spin guard fitted — prevents powered rotation of head/rods when open.	☐	✓	
Drill fluid management	Environmental contamination; spill to water; skin/eye irritation		Drill fluids managed using closed or contained system — returns directed to settlement tanks or IBCs secured to barge. Bunding and geotextile matting used on drill floor. All drilling fluids and cuttings to be removed from barge and disposed of onshore at an approved facility. Marine spill kit including booms available at all times. In the event of a spill, source to be isolated immediately and spill response measures implemented.	☐	✓	
Recovering / handling core and arisings	Strains/sprains; cuts; skin irritation; inhalation		Standard PPE worn. Protective overalls, gloves, and respiratory protection used where contamination is suspected. Core barrel to be laid out horizontally on retrieval. Splits received from the side — no part of the body to be in front of the end of the inner tube during blow-out. Class 2–3 cut-resistant gloves worn when handling rods and core barrel. Wash hands before consuming food.	☐	✓	
Decontamination of equipment	Inhalation, skin irritation, environmental contamination		Standard PPE worn. Wash hands before consuming food. Collect all waste and dispose of at a licensed facility.	☐	✓	
Dropped objects to water	Lost equipment; injury to persons below; environmental impact		Tools and equipment used over or adjacent to water to be secured or tethered where practicable. Certified lifting equipment used for all crane and winch operations. No work beneath suspended loads. Exclusion zones maintained beneath lifting operations and around the drill mast. Barge deck housekeeping maintained to prevent loose items being knocked overboard.	☐	✓	
Barge deck trip hazards from arisings / equipment	Strains, sprains, falls overboard		Offsider to frequently clear arisings with a shovel and maintain clear deck. All equipment to be stowed when not in use.	☐	✓	
BOREHOLE COMPLETION & REINSTATEMENT						

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Borehole grouting (cement / bentonite)	Skin irritation; burns; inhalation		Standard PPE worn including gloves, overalls, safety glasses and dust mask. Open and pour cement in non-windy conditions. Cement sock used to limit dispersal to water. Flush borehole clean of drill cuttings prior to grouting. Grout pumped down to fill borehole — 2% bentonite cement slurry or as specified.	☐	✓	
MANUAL HANDLING						
Twisted, stooped, awkward, asymmetrical postures	Serious back injuries; musculoskeletal disorders		Training in correct lifting techniques.	☐	✓	
Fixed, sustained, rigid, prolonged postures	Serious back injuries; musculoskeletal disorders		Training in correct lifting techniques.	☐	✓	
Unvaried, repetitive movements	Serious back injuries; musculoskeletal disorders		Variation of tasks, rest periods.	☐	✓	
Sudden uncontrolled or jerky movements	Serious back injuries; musculoskeletal disorders; acute strains/sprains		Training in correct lifting techniques. Job/task awareness.	☐	✓	
Using high or sustained force	Serious back injuries; musculoskeletal disorders; strains/sprains; slips/trips/falls		Use lifting aids and hydraulic lifting cables wherever possible. Where equipment cannot be used, two-person lift to be implemented.	☐	✓	
Handling heavy or awkward loads	Serious back injuries; musculoskeletal disorders; strains/sprains		Use lifting aids and hydraulic lifting cables wherever possible. Where equipment cannot be used, two-person lift to be implemented.	☐	✓	
Whole body vibration	Serious back injuries; musculoskeletal disorders		Regular rest periods to be taken.	☐	✓	
Handling that goes on for too long without a break	Acute strains/sprains		Regular rest periods to be taken.	☐	✓	

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LOAD						
Heavy loads	Serious back injuries; musculoskeletal disorders; strains/sprains; slips/trips/falls		Training in the use of hoists and lifting equipment as required. Training as per induction.	☐	✓	
Bulky, unwieldy, unpredictable or uneven loads	Strains/sprains		Use lifting aids and hydraulic lifting cables wherever possible. Two-person lift where equipment cannot be used.	☐	✓	
Unstable, unbalanced, difficult to grip, greasy or slippery loads	Strains/sprains		Use lifting aids and hydraulic lifting cables wherever possible. Two-person lift where equipment cannot be used.	☐	✓	
Hindering the person's view	Strains/sprains		Training in correct loading technique.	☐	✓	
Very hot or cold loads, or loads containing hazardous material	Burns; crushing		Use lifting aids and hydraulic lifting cables wherever possible. Personal protective and correct handling equipment must be used.	☐	✓	
Sharp edges and cutting PVC	Lacerations; strains		Correct PPE (gloves) and lifting devices to be used. Ensure PVC is securely held in place at ergonomic height for cutting.	☐	✓	
ENVIRONMENT						
Slippery, wet, or exposed marine environment	Slips, trips, falls; vessel instability		Ongoing monitoring of weather and sea state. Works suspended when conditions exceed safe operational limits. Non-slip surfaces maintained on barge deck. Wet weather gear supplied and worn as required.	☐	✓	
Work carried out outside in marine environment	Equipment difficult or unsafe to handle/operate		Wet weather gear and environmental PPE supplied and worn as required.	☐	✓	
Polluted or dusty environment	Respiratory difficulties		Personal protective equipment provided and must be worn. Respiratory protection used where contamination is suspected.	☐	✓	
Noisy environment	Hearing loss		Pre-employment hearing test and re-tested every 3 years. Hearing protection supplied (grade 5).	☐	✓	
PEOPLE						

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Insufficiently fit or able to carry out the tasks	Strains/sprains		All potential staff complete a full medical prior to the offer of employment. Regular discomfort and injury assessment compiled to track health/fitness status and enable appropriate allocation of tasks.	☐	☒	
Wearing clothing or footwear that might compromise safe handling	Strains/sprains; crushing		Personal protective equipment supplied. Ongoing checks of PPE condition and compliance.	☒	☐	
Work by themselves	Strains/sprains; no availability of aid		Work teams to comprise two or more people at all times. Management responsible for organising and monitoring daily. On a barge, a minimum two-person crew is mandatory.	☐	☒	
Fatigued by the task	Strains/sprains		Scheduled breaks taken as per the Employment Act. Rest periods taken as required.	☒	☐	
COMMON TASK						
Large horizontal or vertical reaches	Serious back injuries; musculoskeletal disorders; strains/sprains		Lifting equipment or team lift to be employed. Training in the use of hoists/lifting equipment as required.	☐	☒	
Reaching above shoulder height or below mid-thigh	Strains/sprains		Lifting equipment or team lift to be employed. Training in the use of hoists/lifting equipment as required.	☐	☒	
Handling over long distances	Strains/sprains		Lifting equipment or team lift to be employed. Training in the use of hoists/lifting equipment as required.	☐	☒	
Repetitive handling without adequate rest breaks	Serious back injuries; musculoskeletal disorders; strains/sprains		Lifting equipment or team lift to be employed. Rest breaks taken regularly as required.	☐	☒	
Awkward, twisted, or restrained postures	Strains/sprains		Correct lifting techniques and lifting gear to be utilised.	☐	☒	
Standing for a long time	Musculoskeletal disorders		Appropriate rest breaks and pauses as required.	☐	☒	
Prolonged or repetitive squatting, kneeling, or crouching	Musculoskeletal disorders; strains/sprains		Appropriate rest breaks and pauses as required. Task/job share.	☐	☒	
Using poorly designed hand tools	Strains/sprains; lacerations		Specific and specialised tools to be used for each task. Rig and support vessel tool checklist reviewed monthly.	☒	☐	

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ENVIRONMENTAL CONTROLS						
Fuel or drilling fluid spill to water	Environmental contamination; regulatory breach; reputational damage		Bunded fuel storage used at all times on barge. Closed fluid systems used where practicable — returns directed to settlement tanks or IBCs. Marine spill kit including booms available on barge at all times. In the event of a spill, source to be isolated immediately, water pump turned off, and spill response implemented. All spills to be reported. Naturally occurring or environmentally approved drilling fluids to be used where specified.	☐	✓	
Barge bilge water management	Environmental contamination		Bilge water checked for contamination prior to pumping. Contaminated bilge water to be removed to shore for disposal at an approved facility. Spill kit available.	☐	✓	
EMERGENCY EVENTS						
Man overboard	Drowning; hypothermia; fatal injury		Man-overboard procedures communicated and practised at commencement of marine operations. Rescue equipment (throw lines, life rings) immediately accessible on barge at all times. All personnel wearing PFDs at all times on or adjacent to water. Barge master to coordinate response. Emergency services contact details posted on barge.	☐	✓	
Medical emergency on barge	Serious injury or death		At least one trained first aider present during all marine operations. First aid kit available on barge. Emergency contact numbers and evacuation arrangements communicated to all personnel prior to commencement. Access for emergency services considered as part of project planning — to be captured in SSSP.	☐	✓	
Fire on barge	Burns; explosion; vessel loss; drowning		Fire extinguishers available on barge, inspected, and tagged. Ignition sources controlled — particularly near fuel storage and during refuelling. Emergency procedures for fire communicated to all personnel. Evacuation plan includes assembly point and, as last resort, controlled entry to water with PFDs worn.	☐	✓	

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Sign below and/or confirm on SSSP and PSI that you have read and understood the above JSA (also available on phone app QuipCheck):

Name:

Signature:

Date:
